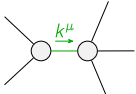
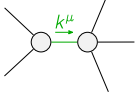
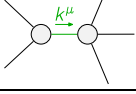
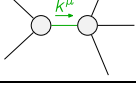
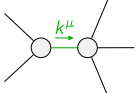
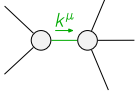
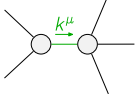
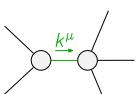


	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1}$	$\alpha\beta\chi$			
$\mathcal{K}_{0^+}^{\#1} \dagger$	$-\frac{3\gamma_1 k^2}{2}$	0				
$\mathcal{K}_{0^-}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{K}_{1^+}^{\#1}$	$\mathcal{K}_{1^+}^{\#2}$		
			$\alpha\beta$	$\alpha\beta$		
			α	α		
	$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$-\gamma_2 k^2$	$-\frac{\gamma_2 k^2}{\sqrt{2}}$	0	0	
	$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$-\frac{\gamma_2 k^2}{\sqrt{2}}$	$-\frac{\gamma_2 k^2}{2}$	0	0	
	$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha}$	0	0	$\frac{\gamma_3 k^2}{2}$	$\frac{\gamma_3 k^2}{\sqrt{2}}$	
	$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha}$	0	0	$\frac{\gamma_3 k^2}{\sqrt{2}}$	$\gamma_3 k^2$	
				$\mathcal{K}_{2^+}^{\#1}$	$\mathcal{K}_{2^-}^{\#1}$	
				$\alpha\beta$	$\alpha\beta\chi$	
				$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$3\gamma_4 k^2$	0
				$\mathcal{K}_{2^-}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{3k^2 \kappa_{15}^{(4)}}{2}$

	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^-}^{\#1}$	$\alpha\beta\chi$						
$\mathcal{T}_{0^+}^{\#1} \dagger$	$\frac{1}{\text{Det}(0^+)}$	0							
$\mathcal{T}_{0^-}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{T}_{1^+}^{\#1} \alpha\beta$	$\mathcal{T}_{1^+}^{\#2} \alpha\beta$	$\mathcal{T}_{1^-}^{\#1} \alpha$	$\mathcal{T}_{1^-}^{\#2} \alpha$			
	$\mathcal{T}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{2}{3 \text{Det}(1^+)}$	$\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	0	0				
	$\mathcal{T}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	$\frac{1}{3 \text{Det}(1^+)}$	0	0				
	$\mathcal{T}_{1^-}^{\#1} \dagger^{\alpha}$	0	0	$\frac{1}{3 \text{Det}(1^-)}$	$\frac{\sqrt{2}}{3 \text{Det}(1^-)}$				
	$\mathcal{T}_{1^-}^{\#2} \dagger^{\alpha}$	0	0	$\frac{\sqrt{2}}{3 \text{Det}(1^-)}$	$\frac{2}{3 \text{Det}(1^-)}$				
						$\mathcal{T}_{2^+}^{\#1} \alpha\beta$	$\mathcal{T}_{2^-}^{\#1} \alpha\beta\chi$		
						$\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$	0	
						$\mathcal{T}_{2^-}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{2}{3 k^2} \kappa_{15}^{(4)}$	

Abbreviations used in matrices

$$\begin{aligned} \gamma_1 &== \kappa_1^{(4)} - 10 \kappa_{15}^{(4)} + 7 \kappa_2^{(4)} \ \&\& \ \gamma_2 == 13 \kappa_{15}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)} \ \&\& \\ \gamma_3 &== 3 \kappa_{15}^{(4)} - 2 \kappa_2^{(4)} \ \&\& \ \gamma_4 == 5 \kappa_{15}^{(4)} - 3 \kappa_2^{(4)} \ \&\& \ \text{Det}(0^+) == -\frac{3}{2} k^2 (\kappa_1^{(4)} - 10 \kappa_{15}^{(4)} + 7 \kappa_2^{(4)}) \ \&\& \\ \text{Det}(1^+) &== -\frac{3}{2} k^2 (13 \kappa_{15}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)}) \ \&\& \ \text{Det}(2^+) == 3 k^2 (5 \kappa_{15}^{(4)} - 3 \kappa_2^{(4)}) \end{aligned}$$

Lagrangian			
$\begin{aligned} &\kappa_1^{(4)} \partial_\beta \mathcal{K}_{\chi\delta}^\delta \partial^{\chi} \mathcal{K}_\alpha^{\alpha\beta} + \kappa_2^{(4)} \partial_\chi \mathcal{K}_{\beta\delta}^\delta \partial^{\chi} \mathcal{K}_\alpha^{\alpha\beta} + \kappa_3^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta - 24 \kappa_{15}^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \\ &18 \kappa_2^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - 2 \kappa_3^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - 27 \kappa_{15}^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta + 18 \kappa_2^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - \\ &\kappa_3^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta + 6 \kappa_{15}^{(4)} \partial^{\chi} \mathcal{K}_\alpha^{\alpha\beta} \partial_\delta \mathcal{K}_{\chi\beta}^\delta - 6 \kappa_2^{(4)} \partial^{\chi} \mathcal{K}_\alpha^{\alpha\beta} \partial_\delta \mathcal{K}_{\chi\beta}^\delta - \frac{15}{2} \kappa_{15}^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \\ &\frac{9}{2} \kappa_2^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - \frac{1}{2} \kappa_3^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \kappa_{15}^{(4)} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \kappa_{15}^{(4)} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$			
Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^-}^{\#1\alpha\beta\chi} == 0$	1	$\begin{aligned} &\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} == \\ &\partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} \end{aligned}$	
$\mathcal{J}_{1^+}^{\#1\alpha} == \mathcal{J}_{1^+}^{\#2\alpha}$	3	$\partial_\chi \partial^\alpha \mathcal{J}^{\beta\chi\alpha} + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha\chi} == 0$	
$\mathcal{J}_{1^+}^{\#1\alpha\beta} == \mathcal{J}_{1^+}^{\#2\alpha\beta}$	3	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} + 2 \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} == 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$	
Total # constraint(s):	7		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$\frac{-11 \kappa_{15}^{(4)} + 6 \kappa_2^{(4)}}{\kappa_{15}^{(4)} (5 \kappa_{15}^{(4)} - 3 \kappa_2^{(4)})}$
	1	0	$\frac{11 \kappa_{15}^{(4)} - 6 \kappa_2^{(4)}}{5 \kappa_{15}^{(4)} \kappa_2^{(4)} - 3 \kappa_{15}^{(4)} \kappa_2^{(4)2}}$
	1	0	$\frac{-156 \kappa_{15}^{(4)3} + 4 \kappa_{15}^{(4)2} (53 \kappa_2^{(4)} - 3 \kappa_3^{(4)}) + 8 \kappa_{15}^{(4)} \kappa_2^{(4)} (-9 \kappa_2^{(4)} + \kappa_3^{(4)}) - \sqrt{3} \sqrt{\kappa_{15}^{(4)2} (339 \kappa_{15}^{(4)2} - 416 \kappa_{15}^{(4)} \kappa_2^{(4)} + 128 \kappa_2^{(4)2}) (13 \kappa_{15}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)})^2}}{\kappa_{15}^{(4)} (5 \kappa_{15}^{(4)} - 3 \kappa_2^{(4)}) (3 \kappa_{15}^{(4)} - 2 \kappa_2^{(4)}) (13 \kappa_{15}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)})}$
	1	0	$\frac{-156 \kappa_{15}^{(4)3} + 4 \kappa_{15}^{(4)2} (53 \kappa_2^{(4)} - 3 \kappa_3^{(4)}) + 8 \kappa_{15}^{(4)} \kappa_2^{(4)} (-9 \kappa_2^{(4)} + \kappa_3^{(4)}) + \sqrt{3} \sqrt{\kappa_{15}^{(4)2} (339 \kappa_{15}^{(4)2} - 416 \kappa_{15}^{(4)} \kappa_2^{(4)} + 128 \kappa_2^{(4)2}) (13 \kappa_{15}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)})^2}}{\kappa_{15}^{(4)} (5 \kappa_{15}^{(4)} - 3 \kappa_2^{(4)}) (3 \kappa_{15}^{(4)} - 2 \kappa_2^{(4)}) (13 \kappa_{15}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)})}$
	1	0	$\begin{aligned} &(156 \kappa_{15}^{(4)3} - 36 \kappa_2^{(4)3} + 2 \kappa_{15}^{(4)} \kappa_2^{(4)} (89 \kappa_2^{(4)} - 7 \kappa_3^{(4)}) + 4 \kappa_2^{(4)2} \kappa_3^{(4)} - \\ &\sqrt{(3 \kappa_{15}^{(4)} - 2 \kappa_2^{(4)})^2 (169 \kappa_{15}^{(4)2} - 200 \kappa_{15}^{(4)} \kappa_2^{(4)} + 60 \kappa_2^{(4)2}) (13 \kappa_{15}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)})^2} + \kappa_{15}^{(4)2} (-290 \kappa_2^{(4)} + 12 \kappa_3^{(4)})) / \\ &(\kappa_{15}^{(4)} (5 \kappa_{15}^{(4)} - 3 \kappa_2^{(4)}) (3 \kappa_{15}^{(4)} - 2 \kappa_2^{(4)}) (13 \kappa_{15}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)})) \end{aligned}$
	1	0	$\begin{aligned} &(156 \kappa_{15}^{(4)3} - 36 \kappa_2^{(4)3} + 2 \kappa_{15}^{(4)} \kappa_2^{(4)} (89 \kappa_2^{(4)} - 7 \kappa_3^{(4)}) + 4 \kappa_2^{(4)2} \kappa_3^{(4)} + \\ &\sqrt{(3 \kappa_{15}^{(4)} - 2 \kappa_2^{(4)})^2 (169 \kappa_{15}^{(4)2} - 200 \kappa_{15}^{(4)} \kappa_2^{(4)} + 60 \kappa_2^{(4)2}) (13 \kappa_{15}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)})^2} + \kappa_{15}^{(4)2} (-290 \kappa_2^{(4)} + 12 \kappa_3^{(4)})) / \\ &(\kappa_{15}^{(4)} (5 \kappa_{15}^{(4)} - 3 \kappa_2^{(4)}) (3 \kappa_{15}^{(4)} - 2 \kappa_2^{(4)}) (13 \kappa_{15}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)})) \end{aligned}$
	2	0	$\begin{aligned} &(-826 \kappa_{15}^{(4)3} + 171 \kappa_2^{(4)3} + 4 \kappa_{15}^{(4)2} (366 \kappa_2^{(4)} - 13 \kappa_3^{(4)}) - 19 \kappa_2^{(4)2} \kappa_3^{(4)} + \kappa_{15}^{(4)} \kappa_2^{(4)} (-865 \kappa_2^{(4)} + 62 \kappa_3^{(4)}) - \\ &\sqrt{(289156 \kappa_{15}^{(4)6} - 16 \kappa_{15}^{(4)5} (73191 \kappa_2^{(4)} - 3479 \kappa_3^{(4)}) + 361 \kappa_2^{(4)4} (-9 \kappa_2^{(4)} + \kappa_3^{(4)})^2 - 62 \kappa_{15}^{(4)} \kappa_2^{(4)3} \\ &(4113 \kappa_2^{(4)2} - 799 \kappa_2^{(4)} \kappa_3^{(4)} + 38 \kappa_3^{(4)2}) + 8 \kappa_{15}^{(4)4} (248467 \kappa_2^{(4)2} - 22457 \kappa_2^{(4)} \kappa_3^{(4)} + 338 \kappa_3^{(4)2}) - 4 \kappa_{15}^{(4)3} \kappa_2^{(4)} \\ &(452001 \kappa_2^{(4)2} - 58234 \kappa_2^{(4)} \kappa_3^{(4)} + 1612 \kappa_3^{(4)2}) + \kappa_{15}^{(4)2} \kappa_2^{(4)2} (928477 \kappa_2^{(4)2} - 151624 \kappa_2^{(4)} \kappa_3^{(4)} + 5820 \kappa_3^{(4)2}))) / \\ &(\kappa_{15}^{(4)} (5 \kappa_{15}^{(4)} - 3 \kappa_2^{(4)}) (3 \kappa_{15}^{(4)} - 2 \kappa_2^{(4)}) (13 \kappa_{15}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)})) \end{aligned}$
	2	0	$\begin{aligned} &(-826 \kappa_{15}^{(4)3} + 171 \kappa_2^{(4)3} + 4 \kappa_{15}^{(4)2} (366 \kappa_2^{(4)} - 13 \kappa_3^{(4)}) - 19 \kappa_2^{(4)2} \kappa_3^{(4)} + \kappa_{15}^{(4)} \kappa_2^{(4)} (-865 \kappa_2^{(4)} + 62 \kappa_3^{(4)}) + \\ &\sqrt{(289156 \kappa_{15}^{(4)6} - 16 \kappa_{15}^{(4)5} (73191 \kappa_2^{(4)} - 3479 \kappa_3^{(4)}) + 361 \kappa_2^{(4)4} (-9 \kappa_2^{(4)} + \kappa_3^{(4)})^2 - 62 \kappa_{15}^{(4)} \kappa_2^{(4)3} \\ &(4113 \kappa_2^{(4)2} - 799 \kappa_2^{(4)} \kappa_3^{(4)} + 38 \kappa_3^{(4)2}) + 8 \kappa_{15}^{(4)4} (248467 \kappa_2^{(4)2} - 22457 \kappa_2^{(4)} \kappa_3^{(4)} + 338 \kappa_3^{(4)2}) - 4 \kappa_{15}^{(4)3} \kappa_2^{(4)} \\ &(452001 \kappa_2^{(4)2} - 58234 \kappa_2^{(4)} \kappa_3^{(4)} + 1612 \kappa_3^{(4)2}) + \kappa_{15}^{(4)2} \kappa_2^{(4)2} (928477 \kappa_2^{(4)2} - 151624 \kappa_2^{(4)} \kappa_3^{(4)} + 5820 \kappa_3^{(4)2}))) / \\ &(\kappa_{15}^{(4)} (5 \kappa_{15}^{(4)} - 3 \kappa_2^{(4)}) (3 \kappa_{15}^{(4)} - 2 \kappa_2^{(4)}) (13 \kappa_{15}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)})) \end{aligned}$
Resolved unitarity condition(s):	(Demonstrably impossible)		